

Power BI Project Final Report

Business Information Systems (BIS)

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Abstract

We analyzed the Global EV Market and Charging Infrastructure dataset to find out the relationship between vehicle adoption rates and charging station availability. The main business problem we tried to solve was identifying infrastructure gaps to support future EV growth. Our analysis concluded that while adoption is increasing, the number of stations per vehicle remains a critical bottleneck in several regions.

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1 Dataset Description

Our group selected the **EV Adoption Rates and Charging Infrastructure** project. The dataset was obtained from **Kaggle**. It contains a comprehensive snapshot of global EV market data for the year 2026. The key variables we focused on were **ev_sales_units** (adoption rate), **charging_stations** (infrastructure count), **country** (geographic distribution), and **co2_reduction_mt** (environmental impact).

2 Data Cleaning (Power Query/ETL)

Before building the dashboard, we cleaned the data in Power Query. The main steps taken were:

- Removed null and blank rows in the 'year' and 'country' columns.
- Created a custom date column using the formula `Date.FromText(Text.From([year]) & "-01-01")` to convert year integers into a Date format.
- Changed the 'country' data category to **Country/Region** to enable accurate map visualizations.
- Removed irrelevant columns that did not contribute to the business KPIs.

3 Data Modeling & DAX

We created a relational data model utilizing a star schema, connecting the main data table to a dedicated Calendar table. To perform our calculations, we wrote several DAX measures.

Listing 1: Total EV Sales Measure

```
ev_sales_units_total = SUM(global_ev_market_charging_infrastructure_2026 [ev_sales_units])
```

Listing 2: Infrastructure Efficiency Ratio

```
EVs_per_Station = DIVIDE([ev_sales_units_total], [TotalChargingStations])
```

Listing 3: Date Table Generation

```
Calendar = CALENDARAUTO()
```

4 Dashboard & Business Insights

Below is the final dashboard created using Power BI, featuring KPI cards for executive overview, a global map for geographic distribution, and brand-share analysis.

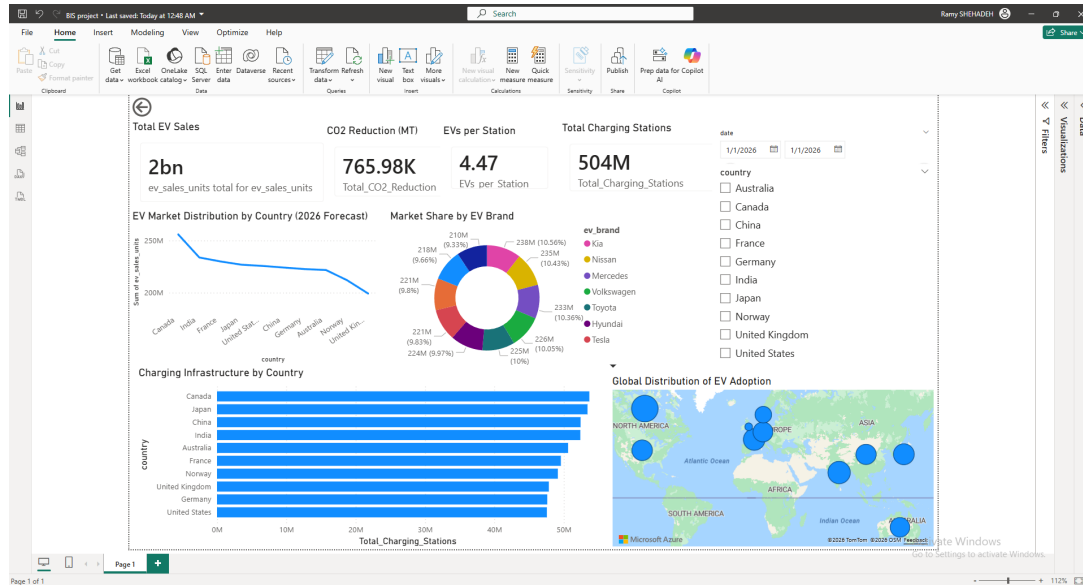


Figure 1: Executive Overview Dashboard

4.1 Key Business Insights found

1. **Infrastructure Gap:** The "EVs per Station" ratio indicates that charging infrastructure is not keeping pace with vehicle sales, suggesting a high risk of "charger anxiety" for users.
2. **Market Dominance:** The brand analysis shows a high concentration of market share among a few top brands, suggesting a need for more competitive entry in the market.
3. **Geographic Concentration:** The map visualization reveals that EV adoption is heavily concentrated in specific global hubs, while other regions remain significantly underserved.

5 Conclusion & Recommendations

Based on our data analysis, we recommend that the company/government should:

- **Prioritize Fast-Charging Networks:** Increase the installation of high-speed chargers in regions with the highest "EVs per Station" ratio.
- **Incentivize Market Entry:** Provide subsidies for emerging EV brands to enter underserved markets to lower prices and increase adoption.
- **Environmental Scaling:** Use the CO2 reduction data to justify further government funding for green infrastructure.

This project demonstrated how raw data can be turned into actionable business decisions using Power BI.